

Interpretation of Policy Functions with Costly External Finance

PS8 Write-up

1 Optimal Capital Policy $K'(K)$ for Low vs. High Productivity

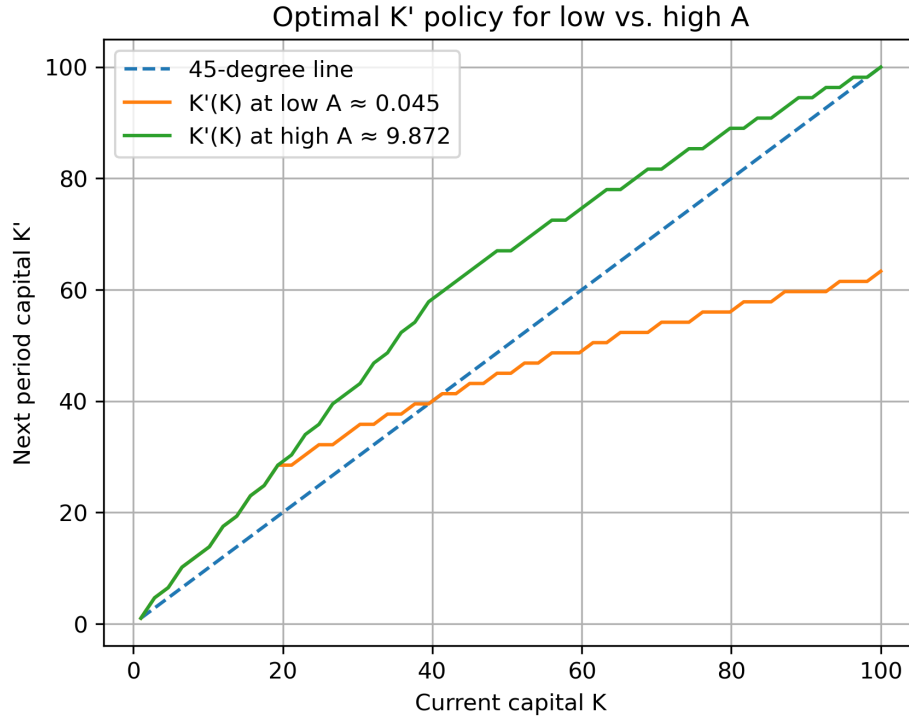


Figure 1: Optimal $K'(K)$ for low and high productivity A .

Figure 1 plots the optimal next-period capital K' as a function of current capital K for a low productivity state and a high productivity state. The dashed 45-degree line represents the benchmark $K' = K$, where the firm keeps its capital stock constant. For low productivity (orange line), the policy lies below the 45-degree line for most of the grid, especially at larger K , indicating that low- A firms eventually prefer to shrink their capital stock rather than keep expanding. For high productivity (green line), $K'(K)$ lies well above the 45-degree line for small and medium K , so high- A firms invest aggressively and grow fast when they are small. As K becomes large, the green line gradually flattens and approaches the 45-degree line, showing that even high-productivity firms reach a size where the marginal value of extra capital falls and further expansion is limited.

2 External Finance Decision by K for Low vs. High Productivity

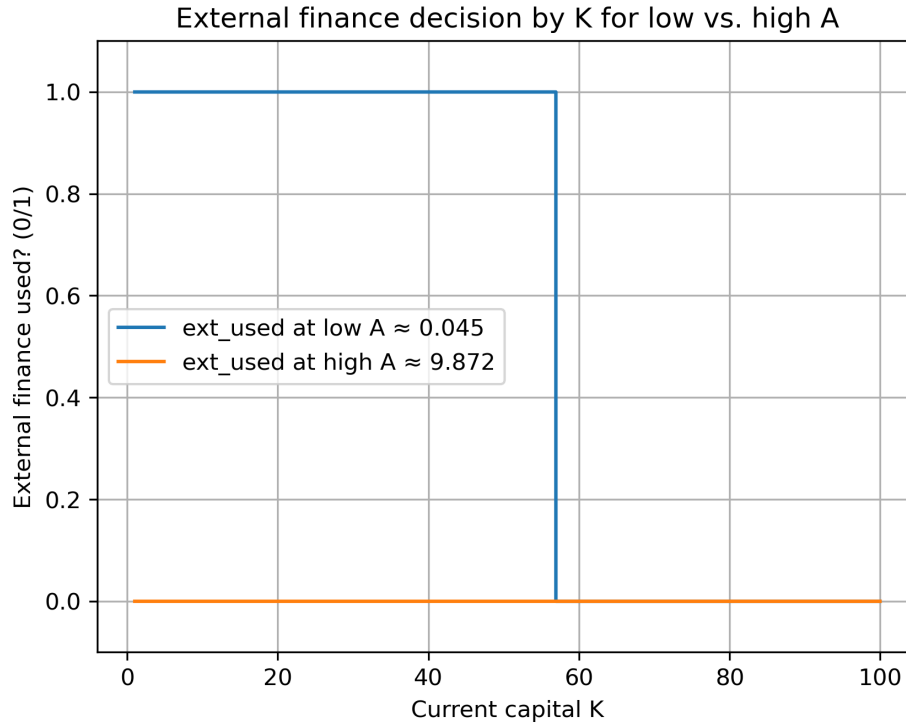


Figure 2: External finance use by capital K for low and high productivity A .

Figure 2 reports, for each level of current capital K , whether the firm uses external finance (value 1) or not (value 0). For low productivity (blue line), the firm relies on external funds for a wide range of capital levels, roughly from very small K up to a middle range of the grid, and then stops using external finance once K is sufficiently high. This pattern is consistent with low- A firms being short of internal funds and needing to borrow when they try to grow, while larger low- A firms can eventually finance investment internally. For high productivity (orange line), the firm almost never uses external finance: strong profits allow it to pay for investment entirely out of current cash flow. Overall, the figure highlights the key mechanism of the model: external finance is mainly used by low-productivity, financially constrained firms, not by high-productivity firms.

3 Optimal Investment Rate $i(K)$ for Low vs. High Productivity

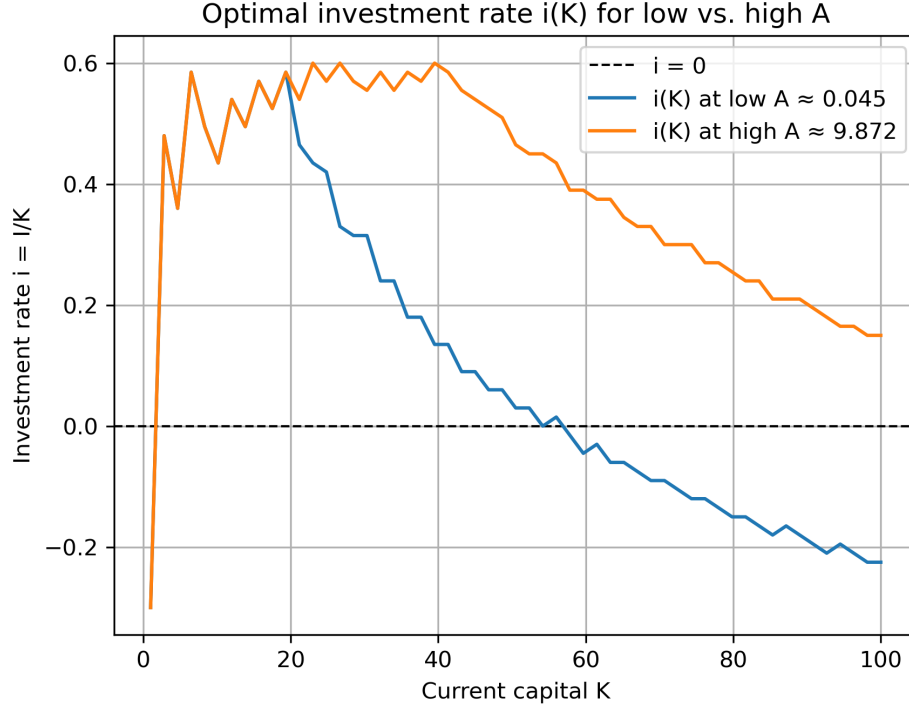


Figure 3: Optimal investment rate $i(K) = I/K$ for low and high productivity A .

Figure 3 shows the optimal investment rate $i(K) = I/K$ as a function of current capital K for low and high productivity states. For high productivity (orange line), the firm chooses very high investment rates at small K (around 0.4–0.6 of the capital stock), which implies rapid growth when the firm is both small and productive; as K rises, the investment rate declines but remains positive, so the firm continues to grow, albeit at a slower pace. For low productivity (blue line), the investment rate is positive at small K but falls much more quickly as the firm becomes larger, eventually crossing zero and becoming negative, which corresponds to disinvestment or letting capital depreciate. The horizontal dashed line at $i = 0$ marks the boundary between expansion and contraction: being above it means net investment, being below it means shrinking the capital stock. Taken together, the figure shows that productivity and size jointly shape investment behavior: high- A firms invest aggressively when small and taper off as they grow, whereas low- A firms invest modestly when small but prefer to run down their capital when they are already large.

4 Main Results

Table 1: Structural estimates and moments (Table 3)

(a) Structural parameter estimates: costly external finance (Θ)							
Model	α	γ	ρ	σ	ϕ ₀		
Costly	0.6956	0.1331	0.0976	0.8932	0.0000		
(b) Moments by model							
Model	a ₁	a ₂	sc(I/K)	std(π/K)	q̄	Ext. frac.	J(Θ)
Moments	0.0300	0.2400	0.4000	0.2500	3.0000	0.2500	n.a.
Costly	0.0626	0.1912	-0.1461	0.3327	2.8782	0.0299	145475.69